

$m(A) < \delta$, 就有 $\int_A f(x)dx < \varepsilon$. 记 $E_n = E(|f_n - f| \geq \varepsilon/m(E))$. 因为 $f_n \xrightarrow{m} f$, $\exists N \in \mathbb{N}$, 当 $n \geq N$, 有 $m(E_n) < \delta$. 在 $E \setminus E_n$ 上, $f(x) \leq f_n(x) + \varepsilon/m(E)$, 从而有

$$\begin{aligned}\int_E f(x)dx &= \int_{E_n} f(x)dx + \int_{E \setminus E_n} f(x)dx \\ &< \varepsilon + \int_{E \setminus E_n} f_n(x)dx + \varepsilon \leq 2\varepsilon + \int_E f_n(x)dx.\end{aligned}$$

$$\Rightarrow \int_E f(x)dx \leq 2\varepsilon + \lim_{n \rightarrow \infty} \int_E f_n(x)dx.$$

19. 由题设有 $0 \leq f(x) - g(x) \leq h(x) - g(x)$. 若 f 可测, 则由

$$\int_E (h(x) - g(x))dx < \varepsilon$$

推得 $f - g \in L(E)$, 所以 $f = [f - g] + g \in L(E)$. 因此只要证明 $f \in \mathfrak{M}(E)$. 事实上, $f(x) = \overline{\lim}_{n \rightarrow \infty} g_n(x) = \underline{\lim}_{n \rightarrow \infty} h_n(x)$.

20. 取 $E = \underline{\lim}_{n \rightarrow \infty} E_n$ 即可.

21. 用反证法.

22. 验证 $\int_{\mathbb{R}} \sum_{n=1}^{\infty} |n^{-a} f(nx)|dx < \infty$.

23. 由题设知, $x^s |f(x)|, x^t |f(x)| \in L(0, \infty)$. 于是当 $u \in (s, t)$ 时,

$$|x^u f(x)| \leq F(x) = \begin{cases} x^s |f(x)|, & 0 < x < 1, \\ x^t |f(x)|, & 1 \leq x < \infty. \end{cases}$$

因为 $F \in L(0, \infty)$, 由定理 3.2.7 知, $I(u)$ 存在, 且连续.

25. 记 $E_+ = \mathbb{R}(f > 0)$, $E_- = \mathbb{R}(f < 0)$, 证明 $m(E_+) = 0, m(E_-) = 0$.

26. 记 $f(x) = \sum_{k=1}^{\infty} |a_k| \chi_{E_k}(x)$, $A_m = \{a \leq x \leq b | f(x) > m\}$. 选取 m_0 使得 $m(A_{m_0}) < \delta/2$, 于是 $m(E_k \setminus A_{m_0}) \geq \delta/2$ 成立. 从而

$$\begin{aligned}\frac{\delta}{2} \sum_{k=1}^{\infty} |a_k| &\leq \sum_{k=1}^{\infty} |a_k| m(E_k \setminus A_{m_0}) \\ &= \int_{[a,b] \setminus A_{m_0}} f(x)dx \leq m_0 \int_{[a,b]} dx = m_0(b-a).\end{aligned}$$